

INVENTORY OPTIMIZATION REPORT

Operations Analytics-1: Inventory Optimization Report

Algorithm: Newsvendor Model (Cost Trade-off Analysis)

Target SKU: Chicken Pepper Bbq Pizza (Medium)

Objective: Mathematically determine the optimal daily kitchen preparation target
to maximize expected profit given uncertain demand and perishable inventory.

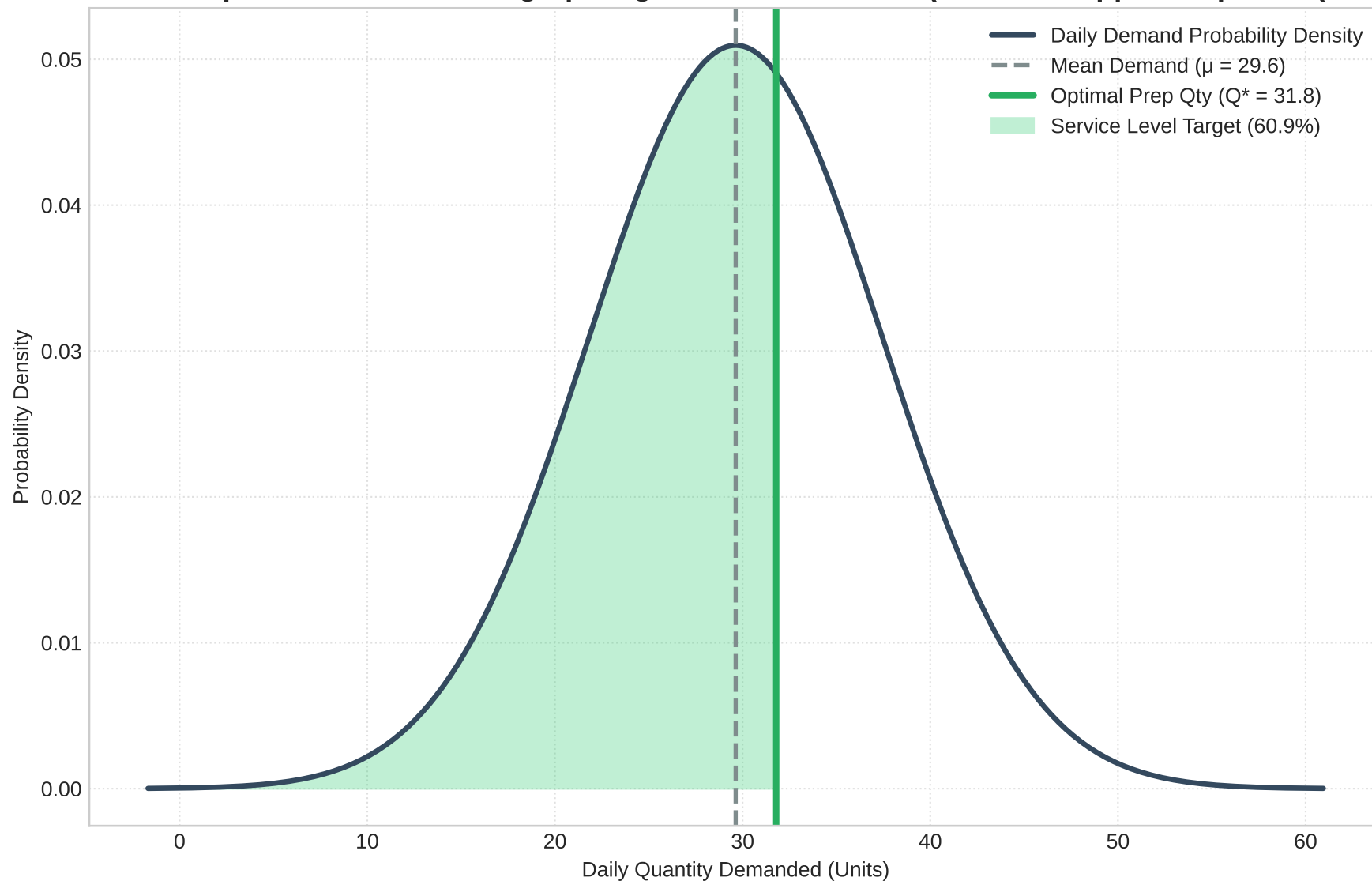
Operational Baselines: Chicken Pepper Bbq Pizza (Medium)

Metric	Value
Unit Sale Price	₹230.00
Unit Prep Cost	₹90.00
Salvage Value	₹0.00
Mean Daily Demand (μ)	29.63 units
Demand Std Dev (σ)	7.83 units

Mathematical Derivation of Optimal Quantity

Economic Parameter	Calculation
Cost of Underage (Cu - Lost Profit)	₹140.00
Cost of Overage (Co - Wasted Cost)	₹90.00
Critical Ratio (Service Level Target)	60.87%
Optimal Z-Score	0.28
Optimal Prep Target (Q*)	31.79 units

Newsvendor Optimization: Balancing Spoilage vs. Lost Revenue (Chicken Pepper Bbq Pizza (Medium))



EXECUTIVE OPERATIONS STRATEGY

Newsvendor Executive Summary:

The critical ratio (CR) dictates that the cost of underage (₹140.00) heavily outweighs the cost of overage (₹90.00). Because profit margins are robust while spoilage costs are relatively contained, conservative preparation destroys value.

Strategic Recommendation:

Despite average daily demand being 30 units, the kitchen operations team MUST prepare 32 units of Chicken Pepper Bbq Pizza (Medium) daily.

By stocking to the 60.9% service level, we mathematically ensure that we capture maximum possible high-margin revenue spikes, accepting that minor daily spoilage is the statistically optimal cost of doing business.